

Design and Development of Efficient Road Object Detection Methods in Extreme Weather Conditions for Secure Autonomous Driving

Project Funding and Investigator Details

- **PI - Dr. Chalavadi Vishnu** Assistant Professor, at the Department of Computer Science and Engineering, IIT Tirupati.
- **Co-PI - Dr. Yeturu Kalidas** Associate Professor, at the Department of Computer Science and Engineering, IIT Tirupati.
- Sanction of grant towards the project titled 'Design and Development of an Efficient Road Object Detection Method in Extreme Weather Conditions for Secure Autonomous Driving' for a duration of 12 months, with a total sanctioned amount of 8,30,500 Rs.
- An amount of 1,50,000 Rs has been received so far on 03/06/2025.

Problems Addressed

- **Extreme Weather Domain Adaptation Architectures** - Proposing architectures using attention mechanisms, especially Transformers and Foundation models that **better understand the contextual features of road objects in various extreme weather conditions** would not only lead to perception systems that will be robust against extreme weather.
- **Reducing Computational Complexity** - Using techniques such as quantization to reduce excessive memory/power usage, to detect objects in an **efficient/optimized manner**.
- **Real-time Defence against Extreme Weather Conditions** - These systems are prone to adversarial attacks either through nature or conscious malicious actors, so safeguarding them through privacy preserving adversarial defence techniques.

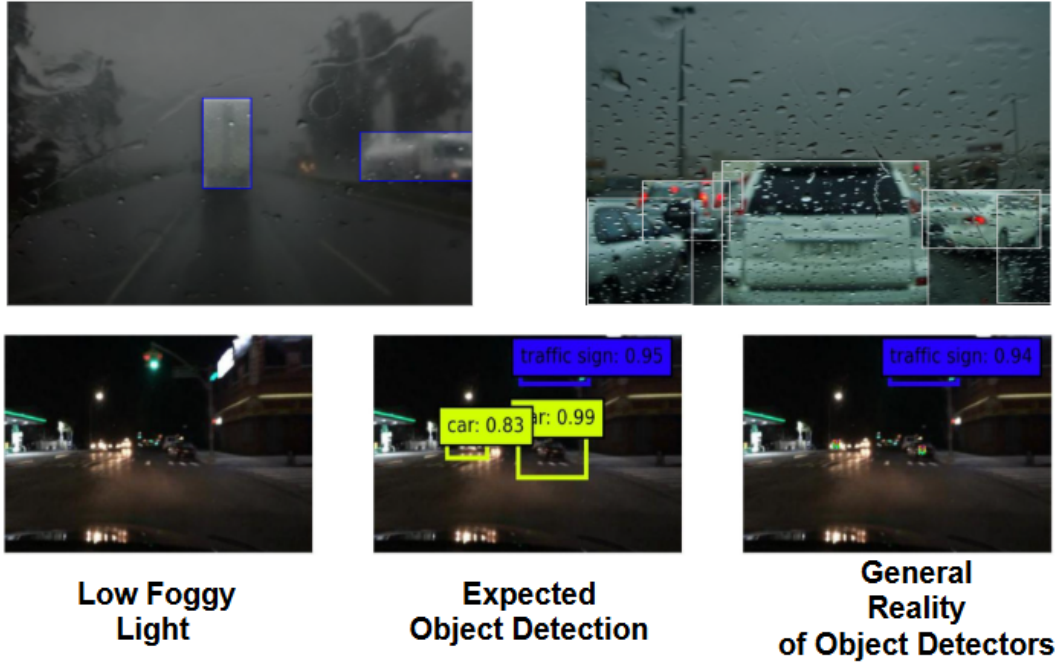


Figure 1: Challenges in extreme weather perception for autonomous driving

Background

- **Extreme Weather Understanding** - Generally, vision and LIDAR sensors in advanced driver assisted systems (ADAS) are modelled from **extreme weather data** (fog, snow, rain, sandstorm) which is quite scarce - **DAWN dataset for example**, often requiring considerably more data for the system to understand weather in the environment better.
- **Road Object Detection in Extreme Weather** - Following along the extreme weather, the **objects on the road and their visibility also affects the ADAS sensors** lane detection, and path planning systems by abrupt weather changes.
- **Computationally Intensive** - Whenever the environment changes, **the ADAS sensors have to utilize more resources** due to the selection of deep learning architectures which might help in understanding the extreme weather along with objects in it, but will require **additional computation, memory, and in turn fuel/battery power**.

AIMs/Objectives

1. Domain Adaptive Architectures/Model for Road Object Detection

- Proposing architectures using attention mechanisms, especially **Transformers**

and **Foundation models** to better understand the contextual features in different weather conditions will lead to perception systems that will be robust against extreme weather scenarios. Using **Generative Adversarial Networks (GANs)**, we will be able to do de-raining, de-foggy, de-storming in real-time on such images/videos.

- Currently, transformers and foundation models require a decent amount of both training data, along with using up a lot of GPU memory to train such models. But, when combined with **Differential convolutional neural network, Quantization, and Continual learning**, we will be able to significantly reduce the computational complexity of these architectures.

2. Adversarial Attacks through Data Poisoning/Model vulnerability

- Although, deep learning architectures are robust against regular noise in the data, these extreme weather conditions along with malicious actors will lead to adversarial attacks that fool the autonomous systems into either mis-classifying the road objects or not detecting them at all.
- We will develop architectures to generate these adversarial samples to understand these attacks better using GANs and Stable Diffusion models.
- We will also propose novel defence algorithms against such attacks using Adversarial training and Defensive distillation techniques as a reference.

Proposed Technology / Solution and Technical Challenges

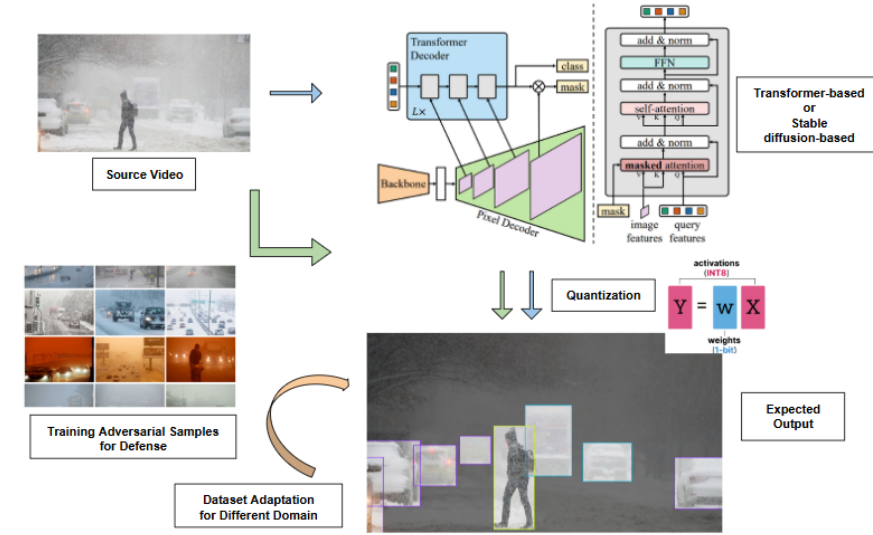


Figure 2: Technical approach for weather-robust perception

TRL:1 (From June to July 2025)

Automatically Removes Extreme Weather Artifacts using Pix2Pix GAN

To develop a Pix2Pix GAN-based deep learning model that automatically removes extreme weather artifacts (rain, snow, smoke, and fog) from road images, enhancing clarity for improved object detection in autonomous driving and surveillance systems. The model will be trained on paired datasets of weather-degraded and clean images, optimized for both high-performance GPUs and edge devices like Raspberry Pi, ensuring robust real-world deployment in adverse conditions.

Model Architecture

- **Framework:** PyTorch
- **GAN Type:** Pix2Pix (Conditional GAN)
- **Generator:** U-Net with skip connections
- **Discriminator:** PatchGAN (patch-level binary classifier)

Training Setup

- **Epochs:** 50
- **Batch Size:** 4
- **Optimizer:** Adam ($\text{lr}=2\text{e-}4$, $\beta=(0.5, 0.999)$)
- **Device:** GPU (CUDA)
- **Result Model:** pix2pix_rain_50epochs.pth

Performance

- **L1 Loss:** 0.0373 (low pixel-wise error)
- **PSNR:** 24.92 dB (high perceptual quality)
- **SSIM:** 0.7646

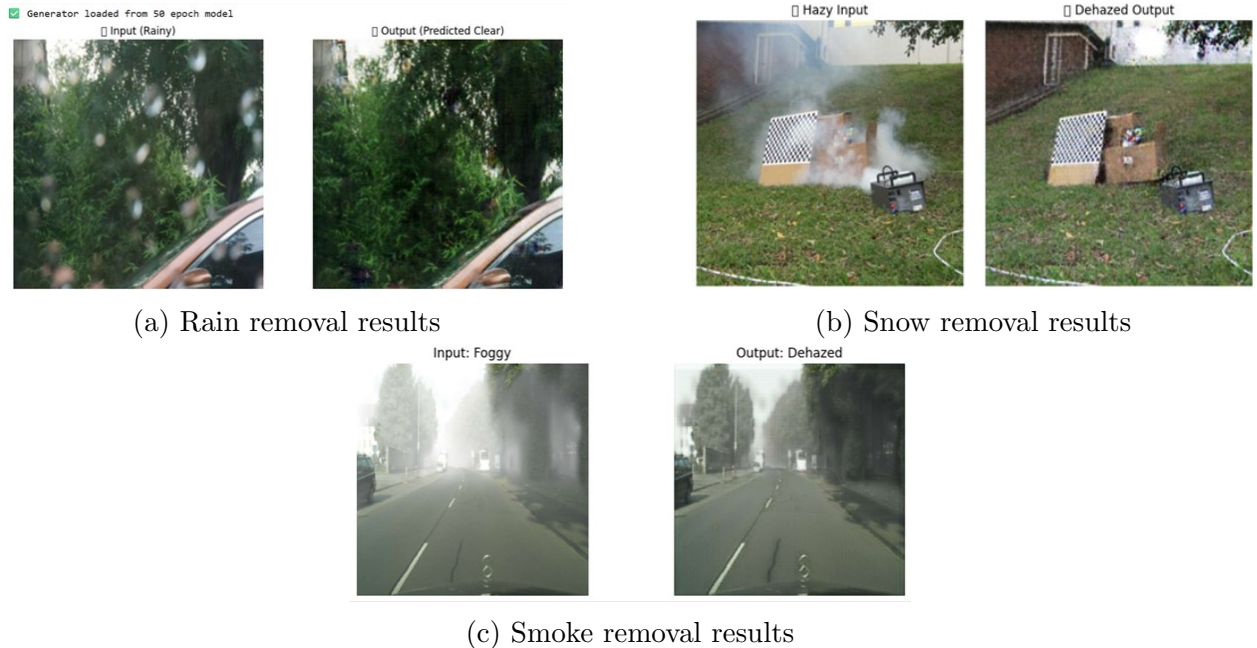


Figure 3: Weather artifact removal outputs

YOLOv5 Object Detection with Enhanced Images

Train a robust YOLOv5-based model to detect road objects after pre-processing images through weather artifact removal GANs.

Model & Training

- **Detector:** YOLOv5s (Ultralytics, small variant)
- **Framework:** PyTorch
- **Input:** 640×640
- **Epochs:** 50
- **AMP:** Mixed-precision enabled

Table 1: Training Metrics for YOLOv5

Metric	Value
Training Time	1.592 Hour
Best Model Path	runs/train/my_model/weights/best.pt
Box Loss (Last Epoch)	0.0338
Objectness Loss	0.0304
Classification Loss	0.0040



Figure 4: YOLOv5 detection results on weather-enhanced images

Conclusion

The goal of this project is to develop a robust object detection pipeline that enhances the accuracy of YOLOv5 in challenging weather conditions commonly encountered in road scenes, such as rain, fog, or snow. To achieve this, the raw input images are first passed through a weather artifact removal model based on Generative Adversarial Networks (GANs), which are trained to clean and restore visibility by removing distortions caused by adverse weather. This pre-processing step significantly improves image clarity and contrast, allowing the YOLOv5 model to better identify and localize critical road objects such as vehicles, pedestrians, traffic signs, and road markings. The overall system is designed to improve the performance and reliability of object detection models in real-world autonomous driving scenarios, where robust perception under varying environmental conditions is essential for safety and decision-making.

TRL:2 (July to August 2025)

1. JetAuto Standard Kit Hardware Procurement

- Initiating the purchase process for the JetAuto robotics platform, including necessary modules such as Jetson Nano, sensors, and optional display units, for research and development in autonomous navigation projects.

2. Advanced Model Training

- Training more advanced models such as YOLOv8, YOLOv11, or transformer-based architectures like DETR instead of YOLOv5, to improve detection accuracy and robustness.

3. Model Optimization

- Apply quantization (converting FP32 to INT8) and pruning techniques to reduce the model size and enhance inference speed, especially for deployment on resource-constrained devices.

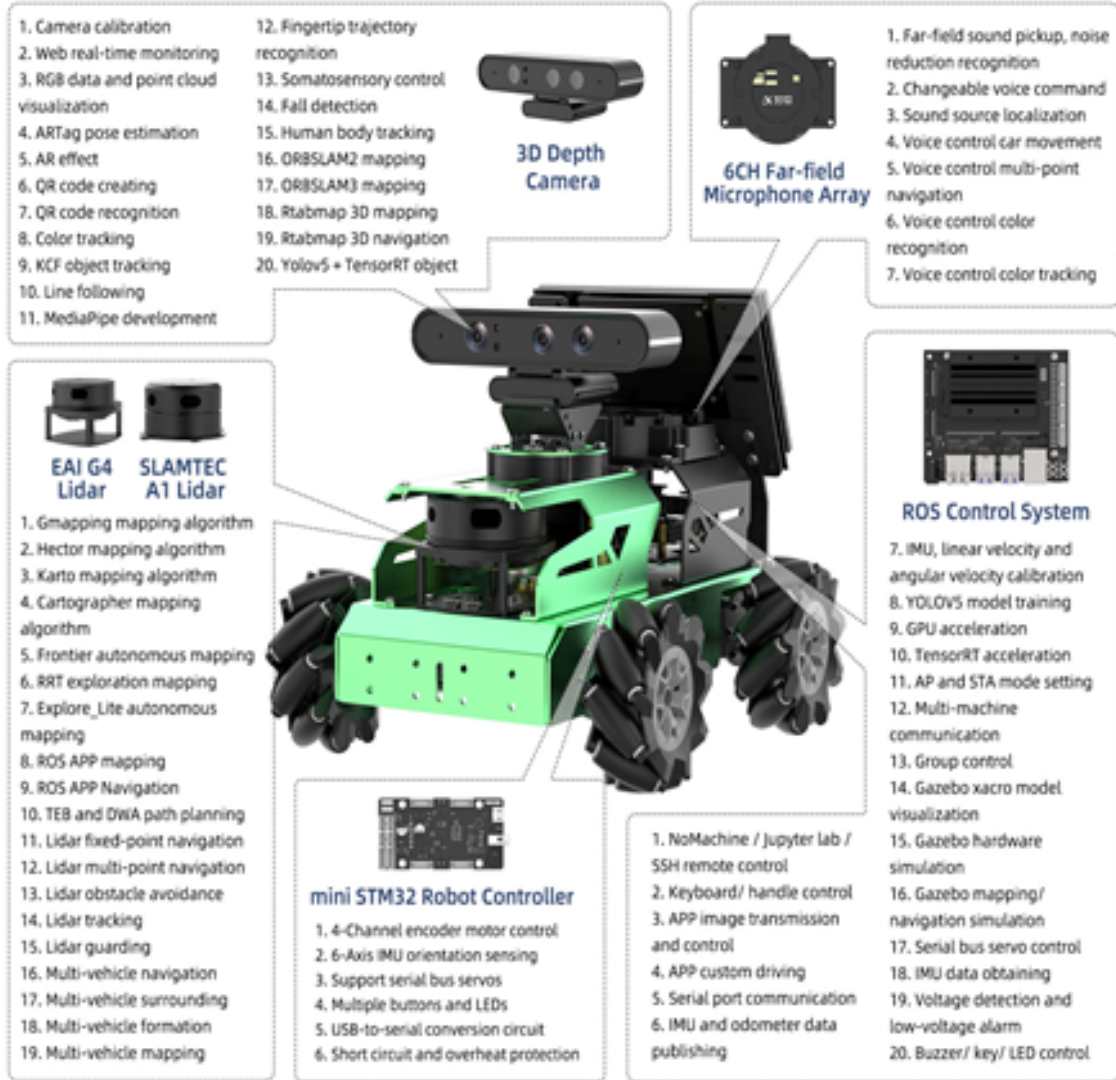


Figure 5: JetAuto Standard Kit for deployment

- Explore hardware acceleration using frameworks like TensorRT or ONNX Runtime to optimize performance on GPUs and TPUs.

4. Enhanced Datasets

- Expand the training dataset by incorporating a wider range of diverse weather conditions such as sandstorms, heavy snowfall, and fog.
- Include night-time and low-light scenarios to improve the model's robustness and performance under challenging visual environments.



Embeddvar Technology Solutions

GSTIN : 33BIEPG4303B1Z6
Contact : 9486806486
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Date :: 22-07-2025

Quote :: 525

Validity :: 30 Days

VENDOR DETAILS

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CLIENT DETAILS

Dr Chalavadi Vishnu
Assistant Professor Dept of CSE
IIT Tirupati, Andhra Pradesh
Tiurpati - 517 619

S.No	Product Descripton	Quantity	Rate/Qty Rs. Ps.	Total Amount
1	JetAuto ROS Robot Standard Kit with Jetson Nano 4 GB and 7 Inch Touch Display	1	101695.00	101695.00
Total ::				101695.00
Others ::				0.00
GST @18% ::				18305.10
Total Amount				120000.00

Warranty : One Year

Terms of Payment :: 100% Payment within 30 Days of Delivery

Terms of Delivery :: C & F from Chennai within 4 Weeks from the receipt of Purchase Order

Our Bank Details ::
Name of Account :: Embeddvar Technology Solutions
Account No :: 510 9090 1011 3890
Type of Account :: Current Account
Name of Bank :: CITY Union Bank, Avadi Branch, Chennai
IFSC No :: CIUB 0000 266
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